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PAPER_1079: Frozen Planet Solar Wind Power — Direct Kinetic Flux Powering of Outer Solar System Bodies

Abstract

Frozen planets beyond the frost line ($d > 5$ AU) receive their core energy primarily from solar wind kinetic flux rather than Ug3 magnetic string coupling. We derive the kinetic energy flux $\Phi_{\text{kinetic}}(d) = \frac{1}{2}\rho_{\text{sw}}(d)v_{\text{sw}}^3$ and compute the power delivered to the cores of Uranus, Neptune, Pluto, and Eris through their magnetospheric cross-sections. Results show core power ranging from $\sim 10^{14}$ W (Uranus) to $\sim 10^3$ W (Eris), validating the UQFF prediction that frozen bodies are powered directly by solar winds.

§1 Kinetic Energy Flux

$$\Phi_{\text{kinetic}}(d) = \frac{1}{2}\rho_{\text{sw}}(d) \cdot v_{\text{sw}}^3$$

with density scaling:

$$\rho_{\text{sw}}(d) = \rho_{\text{sw,1AU}} \cdot \left(\frac{\text{AU}}{d}\right)^2$$

§2 Magnetospheric Capture

$$P_{\text{captured}} = \Phi_{\text{kinetic}}(d) \cdot \pi R_{\text{mag}}^2$$

$$P_{\text{core}} = P_{\text{captured}} \cdot \eta_{\text{penetration}}$$

§3 Frozen Body Results

Body	d (AU)	R_{mag} (m)	Φ_k (W/m^2)	P_{core} (W)
Uranus	19.19	1.8×10^9	1.36×10^{-7}	$\sim 1.4 \times 10^{11}$
Neptune	30.07	2.4×10^9	5.53×10^{-8}	$\sim 1.0 \times 10^{11}$
Pluto	39.48	1.2×10^6	3.22×10^{-8}	$\sim 1.5 \times 10^4$
Eris	67.67	1.0×10^6	1.10×10^{-8}	$\sim 3.4 \times 10^3$

§4 Frost Line Transition

The frost line at $d_{\text{frost}} \approx 5$ AU marks the transition where solar wind kinetic power dominates over Ug3 magnetic string coupling as the primary core energy source. Beyond this distance:

- Ug3 string tension decays as $\cos(\omega_s t)$ with diminishing amplitude

- Solar wind kinetic flux, while falling as d^{-2}, remains the dominant energy input channel for bodies with significant magnetospheres
- Small bodies (Pluto, Eris) with $R_{\rm mag} \sim 10^6$ m receive minimal power, consistent with their geological quiescence

§5 Solar Cycle Variation

- **Solar maximum:** $\rho_{\rm sw} = 1.6 \times 10^{-20}$, $v_{\rm sw} = 8 \times 10^5$ m/s $\to P_{\rm core}$ increases $\sim 8\times$
- **Solar minimum:** $\rho_{\rm sw} = 4 \times 10^{-21}$, $v_{\rm sw} = 3 \times 10^5$ m/s $\to P_{\rm core}$ decreases $\sim 5\times$

References

- PAPER_1078: Solar Wind Flux Partitioning Budget
- PAPER_412: HeliosphereHydrogenComplexSCmStellarAgeCalculator
- Star-Magic.txt: §4 Frozen Planet Wind Power

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic